

Stat 215A Lecture Notes

Lance Remigio

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1 Week 7

1.1 Monday

1.2 Wednesday

Recall that Ω and $\mathcal{C}$ where $\mathcal{C}$ is a set of sets and U is a simply just a set.

Proposition. $\sigma(\mathcal{C} \cap U) = \sigma(\mathcal{C}) \cap U$

Proof. ($\supseteq$) Define the set

$$\mathcal{G} = \{A : A \cap U \in \sigma(\mathcal{C} \cap U)\}.$$

Last time we showed that

(1) $\Omega \in \mathcal{G}$

(2) for every $A \in \mathcal{G}$, $A^c \in \mathcal{G}$.

Now we will show that for a sequence of sets $A_n \in \mathcal{G}$ for all $n \in \mathbb{N}$, their union is contained in the set. Let A_n where $n \in \mathbb{N}$ is a sequence of sets in $\mathcal{G}$. Our goal is to show that

$$\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{G}$$

that is,

$$\bigcup_{n \in \mathbb{N}} A_n \cap U \in \sigma(\mathcal{C} \cap U).$$

Since $A_n \in \mathcal{G}$, we have

$$A_n \cap U \in \sigma(\mathcal{C} \cap U) \quad \forall n \in \mathbb{N}.$$

Since $\mathcal{C} \cap U$ is a sigma algebra, we have that

$$\bigcup_{n \in \mathbb{N}} (A_n \cap U) \in \sigma(\mathcal{C} \cap U).$$

But this tells us that

$$\bigcup_{n \in \mathbb{N}} A_n \cap U \in \sigma(\mathcal{C} \cap U) \implies \bigcup_{n \in \mathbb{N}} A_n \cup U \in \sigma(\mathcal{C} \cap U).$$

Hence, $\bigcup_{n \in \mathbb{N}} A_n \in \sigma(\mathcal{C} \cap U)$. ■

Proposition. $\sigma(\mathcal{C}) \cap U \subseteq \sigma(\mathcal{C} \cap U)$.

Proof. Note that $\mathcal{C} \subseteq \mathcal{G}$. Indeed, if $A \in \mathcal{C}$, then $A \cap U \in \mathcal{C} \cap U \subseteq \sigma(\mathcal{C} \cap U)$. Also, $\sigma(\mathcal{C}) \subseteq \mathcal{G}$ because $\mathcal{C} \subseteq \mathcal{G}$, we have

$$\sigma(\mathcal{C}) \subseteq \sigma(\mathcal{G}) = \mathcal{G}.$$

To show finally that $\sigma(\mathcal{C}) \cap U \subseteq \sigma(\mathcal{C} \cap U)$. Let $B \in \sigma(\mathcal{C}) \cap U$. Hence, $B = A \cap U$ for some $A \in \sigma(\mathcal{C})$. Since $\sigma(\mathcal{C}) \subseteq \mathcal{G}$, $A \cap U \in \sigma(\mathcal{C} \cap U)$. Hence, this tells us that $B \in \sigma(\mathcal{C} \cap U)$. Since B was arbitrary,

$$\sigma(\mathcal{C}) \cap U \subseteq \sigma(\mathcal{C} \cap U).$$

■

Example. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{C} = \{\{1, 3, 5\}, \{2, 4, 6\}\}$ and $u = \{1, 2\}$. Our goal is to show that

$$\sigma(\mathcal{C} \cap u) = \sigma(\mathcal{C}) \cap u.$$

Indeed, notice that

$$(1) \ C_1 \cap u = \{1, 3, 5\} \cap \{1, 2\} = \{1\};$$

$$(2) \ C_2 \cap u = \{2, 4, 6\} \cap \{1, 2\} = \{2\}.$$

Then we have

$$\underbrace{\sigma(\mathcal{C} \cap U)}_{\sigma\text{-algebra on } u} = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

and

$$\underbrace{\sigma(\mathcal{C})}_{\sigma\text{-algebra on } \Omega} = \{\emptyset, \{1, 3, 5\}, \{2, 4, 6\}, \{1, 2, 3, 4, 5, 6\}\}$$

and

$$\sigma(\mathcal{C}) \cap u = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$$

Hence, we see that

$$\sigma(\mathcal{C} \cap u) = \sigma(\mathcal{C}) \cap u.$$

Recall the proof of prop E.1.5 says in its last paragraph

$$\text{"If } A \in \sigma(\mathcal{C}) \implies A \in \mathcal{G}\text{"}$$

This implies that $A \cap U \in \sigma(\mathcal{C} \cap U)$ and so $A \in \sigma(\mathcal{C} \cap U)$. It follows that $\sigma(\mathcal{C}) \cap U \subseteq \sigma(\mathcal{C} \cap U)$.

Example (Simple Counter-Example to the Author's Proof). Let $\Omega = \{1, 2, 3\}$ and $\mathcal{C} = \{\{1\}, \{2\}\}$ and $U = \{2, 3\}$. Then

$$\sigma(\mathcal{C}) = \{\emptyset, \Omega, \{1\}, \{2\}, \{2, 3\}, \{1, 3\}, \{3\}, \{1, 2\}, \Omega\}$$

This is a σ -algebra on Ω . Also,

$$\sigma(\mathcal{C} \cap U) = \sigma(\{\{\emptyset, \{2\}\}\}) = \{\emptyset, \{2\}, \{3\}, \{2, 3\}\}.$$

let $A = \{1\} \in \sigma(\mathcal{C})$. Furthermore, $A \cap U \in \sigma(\mathcal{C} \cap U)$ since $A \cap U = \emptyset$. But note that $A = \{1\} \notin \sigma(\mathcal{C} \cap U)$.

1.3 The Measure Function

Definition (Measure Function). Let $\mathcal{F}$ be an algebra. A **measure function** $\mu : \mathcal{F} \rightarrow (0, \infty) \cup \{\infty\}$ that satisfies

(1) **(Empty set)** $\mu(\emptyset) = 0$

(2) **(Countable Additivity)** If $E_n \in \mathcal{F}$ is a sequence of pairwise disjoint events in $\mathcal{F}$, then

$$\mu\left(\bigcup_{n \in \mathbb{N}} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n).$$

(3) If $\mathcal{F}$ is a σ -algebra, the triplet $(\Omega, \mathcal{F}, \mu)$ is a measure space.

Definition (Probability Measure). A **Probability measure** is a measure that satisfies $\mu(\Omega) = 1$.

1.4 Monday Lecture

Proposition (Continuity of a Measure). Any measure $\mu(\Omega) < \infty$ satisfies the following properties:

(i) **(Finite additivity):** For a finite sequence of disjoint sets $E_1, E_2, \dots, E_k$

$$\mu\left(\sum_{i=1}^k E_i\right) = \sum_{i=1}^k \mu(E_i).$$

Proof. Suppose we have an $\mathcal{F}$ -algebra where $\mu : \mathcal{F} \rightarrow [0, \infty) \cup \{\infty\}$. If E_n where $n \in \mathbb{N}$ is a sequence of disjoint sets; that is, $E_i \cap E_j = \emptyset$ if $i \neq j$. Then

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i).$$

Note that in (i), given the above, we can write

$$\mu\left(\bigcup_{i=1}^k E_i\right) = \mu\left(\sum_{i=1}^k E_i\right) = \sum_{i=1}^k \mu(E_i).$$

This statement can be easily proven using property (1) of the definition of a measure where all indices i past k , we can set $E_i = \emptyset$. Indeed, we will consider three cases:

(1) $i \neq j$ where $i, j \in \{1, 2, \dots, k\}$ implies $E_i \cap E_j = \emptyset$ where $E_1, E_2, \dots, E_k$ are all disjoint.

(2) $i \neq j$ where $i, j \in \{k+1, k+2, \dots\}$ implies $E_i \cap E_j = \emptyset \cap \emptyset = \emptyset$.

(3) $i \in \{1, 2, \dots, k\}$ and $j \in \{k+1, k+2, \dots\}$ implies $E_i \cap E_j = E_i \cap \emptyset = \emptyset$.

From (i)-(iii), we have E_i where $i \in \mathbb{N}$ is a disjoint sequence of sets. The above proposition includes E_n for $n \in \mathbb{N}$ disjoint and

$$\mu\left(\bigcup_{i=1}^{\infty} E_n\right) = \sum_{i=1}^{\infty} \mu(E_n),$$

which implied that $\bigcup_{i=1}^{\infty} E_i \in \mathcal{F}$. Note that because

$$\begin{aligned} \mu\left(\bigcup_{i=1}^{\infty} E_i\right) &= m\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i) \\ &= \sum_{i=1}^k \mu(E_i) + \sum_{i=k+1}^{\infty} \mu(\emptyset) \\ &= \sum_{i=1}^k \mu(E_i) + 0 \\ &= \sum_{i=1}^k \mu(E_i). \end{aligned}$$

■

Recall that if f is continuous, then $x_n \rightarrow x$ which implies that $f(x_n) \rightarrow f(x)$.

Proposition (Measure is Continuous from Below). If E_n is a non-decreasing sequence to E , $E \in \mathcal{F}$ and $E_n \in \mathcal{F}$ for all $n \in \mathbb{N}$, then $\mu(E_n)$ converges to $\mu(E)$ where $\mu(E) \in \{\sup E, \infty\}$.

1.5 Wednesday Lecture

Redo the following propositions on homework.

Proposition (B.2.5 (Redo some of the details in your proof)). $\lim_{x \rightarrow y} f(x) = v$ if and only if $\lim_{n \rightarrow \infty} x^{(n)} = y$ and for all $n \in \mathbb{N}$, $x^{(n)} \neq y$, then

- (*) $\lim_{n \rightarrow \infty} x^{(n)} = y$ where $x^{(n)} \neq y$ for all $n \in \mathbb{N}$ means for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n > N$ ($n \in \mathbb{N}$), $d(x^{(n)}, y) < \varepsilon$.
- (*) If $\lim_{x \rightarrow y} f(x) = v$ means that for all $\varepsilon > 0$, there exists $\delta > 0$ such that if $d_A(x, y) < \delta$, then $d_B(f(x), v) < \varepsilon$.

Proof. ($\implies$) Since $\lim_{x \rightarrow y} f(x) = v$. Let $\varepsilon > 0$ be given. Then there exists $\delta > 0$ such that if $d_A(x, y) < \delta$, then $d_B(f(x), v) < \varepsilon$. Then if $\lim_{n \rightarrow \infty} x^{(n)} = y$ ($x^{(n)} \neq y$) for all $n \in \mathbb{N}$.

Our goal is to show that for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n > N$, $d_B(f(x^{(n)}), v) < \varepsilon$.

Then for the aforementioned $\delta > 0$, there exists $N \in \mathbb{N}$ if $n > N$, $d(x^{(n)}, y) < \delta$, then $d_B(f(x^{(n)}), v) < \varepsilon$. Since ε was arbitrary, for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that if $n > N$, then $d_B(f(x^{(n)}), v) < \varepsilon$. Hence, $\lim_{n \rightarrow \infty} f(x^{(n)}) = v$.

($\impliedby$) Suppose $\lim_{n \rightarrow \infty} x^{(n)} = y$ where ($x^{(n)} \neq y$ for all $n \in \mathbb{N}$) and for sake of contradiction $\lim_{x \rightarrow y} f(x) \neq v$; that is, there exists an $\varepsilon > 0$ such that for all $\delta > 0$, there exists an $x \neq y \in A$ such that if $d_A(x, y) < \delta$, $d_B(f(x), v) \geq \varepsilon$. For our fixed $\varepsilon > 0$, let $\delta = 1$. Then there exists an $x_1 \neq y \in A$ such that if $d_A(x^{(1)}, y) < 1$, then $d_B(f(x_1), v) \geq \varepsilon$. Then for $\delta = \frac{1}{2}$, there exists an $x_2 \neq y \in A$ such that if $d_A(x^{(2)}, y) < \frac{1}{2}$, then $d_B(f(x^{(3)}), v) \geq \varepsilon$. We continue in this manner until, we have a sequence $(x_n) \subseteq A$ where $x_n \neq y$ for all $n \in \mathbb{N}$ where if $d_A(x^{(n)}, y) < \frac{1}{n}$, then $d_B(f(x^{(n)}), v) \geq \varepsilon$. ■

Proposition. Prove that if $d(x^{(n)}, y) < \frac{1}{n}$, then $\lim_{n \rightarrow \infty} x^{(n)} = y$.

Proposition (Redo). Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be continuous functions on A and B , respectively. Then $g \circ f : AC$ is continuous on A .

Proof. Our goal is to show that B_n for $n \in \mathbb{N}$ is a disjoint sequence of sets.

Note that $B_1 = E_1$ (**This is a homework problem**) and $B_n = E_n \setminus E_{n-1}$. Let $i \neq j$ where $i, j \in \{2, 3, \dots\}$. Without loss of generality, let $i < j$. Then our goal is to show that $B_i \cap B_j = \emptyset$. Note that $B_i = E_i \setminus E_{i-1}$ and $B_j = E_j \setminus E_{j-1}$. Then

$$\begin{aligned} B_i \cap B_j &= (E_i \cap E_{j-1}^c) \cap (E_j \cap E_{j-1}^c) \\ &= (E_i \cap E_j) \cap (E_{i-1}^c \cap E_{j-1}^c). \end{aligned} \quad (*)$$

Since $E_i \subseteq E_j$, $E_i \cap E_j = E_i$ and since $E_{i-1} \subseteq E_{j-1}$ implies that $E_{j-1}^c \subseteq E_{i-1}^c$, and so $E_{i-1}^c \cap E_{j-1}^c = E_{j-1}^c$. Then we have (*) implies that $B_i \cap B_j = E_i \cap E_{j-1}^c$. If $x \in E_i \cap E_{j-1}^c$ implies $x \in E_i$ and $x \notin E_{j-1}$ but $E_i \subseteq E_{j-1}$ (**Polish the rest of this for homework**). ■

Proposition ((3) Continuity from Above).